Punctual or Continuous? Analyzing Depression Traces in Language and Paralanguage with Multiple Instance Learning

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Abstract

The key-question addressed in this article is whether the traces of depression in speech are punctual (the pathology manifests itself at specific points in time) or continuous (the pathology manifests itself at every moment in time), where the expression "speech" refers to both speech signals and their transcriptions. For this reason, this work compares the performances of different approaches (both unimodal or multimodal) based either on the assumption that the traces are punctual or on the other one. In this way, it is possible to test which of the alternative assumptions is more realistic. The experiments were performed over a publicly available dataset (the Androids Corpus) and the results include F1 Scores up to 93.1%, among the best reported in literature for the Corpus. Furthermore, the results suggest that depression traces are punctual, but they appear so frequently that approaches based on the assumption of continuous traces still perform well. The conclusions discuss the implications of such observation.

CCS Concepts

• **Computing methodologies** → **Speech recognition**; *Natural language processing*; **Artificial intelligence**.

Keywords

Depression detection, Multiple Instance Learning, Language, Paralanguage

ACM Reference Format:
Rawan Alsarrani, Anna Esposito, and Alessandro Vinciarelli. 2025. Punctual or Continuous? Analyzing Depression Traces in Language and Paralanguage with Multiple Instance Learning. In *Proceedings of the 27th International Conference on Multimodal Interaction (ICMI '25), October 13–17, 2025, Canberra, ACT, Australia.* ACM, New York, NY, USA, 10 pages. https://doi.org/10.1145/3716553.3750809

1 Introduction

Depression is "[...] an increase in negative emotions and feelings and a reduction in positive emotions and feelings [...]" [28], a debilitating pathology leading to "[...] depressed mood, loss of interest or pleasure, decreased energy, feelings of guilt or low self-worth, disturbed sleep or appetite, and poor concentration [...]" [36], possibly over long periods of time (up to several months). The World Health Organization estimates that, in 2015, the number of patients was greater than 300 millions (4.4%of the world's population at the time) and depression was responsible for 7.5% of all years lived with disability [1]. A parallel study, conducted in 29 countries, showed that the consumption of antidepressants doubled, on average, between 2000 and 2015 [18]. Such an increase is not due to depression only (antidepressants treat many other problems), but it still suggests that the number of people affected by the pathology keeps increasing.

Automatic technologies for depression detection can help dealing with the problems above (see [11] for an extensive survey). The key-assumption underlying all proposed approaches is that the pathology leaves physical, machine detectable traces in the way people behave (facial expressions, tone of voice, lexical choices, etc.). For this reason, the literature proposes a wide spectrum of depression traces, often referred to as *markers*, likely to account for the presence of depression. This apply especially to *speech*, where such an expression refers to both the speech signals and their transcriptions (see Section 2) [39]. The reason is that speech is commonly used for depression detection because it can be captured easily and unobtrusively [13], something important when dealing with potential depression patients.

In the context above, this article tries to address a question that, to the best of our knowledge, is still open, namely whether depression traces in speech are *continuous* (they can be detected at every point in time) or *punctual* (they can be detected only at specific points in time). This is important because, if the traces are continuous, depression detection can be performed reliably using a few seconds of speech. This would be a major advantage because depression patients find it difficult to engage in long spoken interactions.

The experiments were performed over a publicly available benchmark (the Androids Corpus [52]) and address the question through the comparison between two approaches. The difference between the two is that the classifier they use is either trained in a standard way, referred to as *Single Instance Learning* (SIL) hereafter, or with a *Multiple Instance Learning* methodology [9]. MIL is a body of approaches designed for weakly labeled data in which an item to be classified (a speech recording in the experiments) gives rise to multiple feature vectors that share the same label, but do not necessarily belong to the same class (e.g., some of the feature vectors extracted from depressed speech share their properties with those extracted from non-depressed speech and vice versa). If an approach based on SIL performs at the same level as one based on MIL, it means that most feature vectors convey reliable depression-relevant information, thus suggesting that the traces are continuous. In contrast, if a MIL-based approach improves over a SIL-based one, it means

ICMI '25, Canberra, ACT, Australia

ACM ISBN 979-8-4007-1499-3/25/10
https://doi.org/10.1145/3716553.3750809

that a few vectors carrying depression-relevant information (the *witness instances* in MIL terminology) can be sufficient to detect the pathology, thus suggesting that depression traces are punctual.

To the best of our knowledge, this is the first work addressing the problem and the main contributions can be summarized as follows:

- First analysis of the way depression traces distribute over speech;
- First attempt to analyze the impact of the information obtained at the previous item over depression detection performance.

The rest of this article is organized as follows: Section 2 surveys previous work, Section 3 describes the methodologies used in the experiments, Section 4 presents experiments and results, and the final Section 5 draws some conclusions.

2 Survey of Previous Work

This section surveys the main features identified as depression markers in speech signals and language, the two modalities used in the experiments of this work.

2.1 Depression Traces in Speech Signals

According to a survey, "many studies have documented the relationship between objective acoustical measures of voice and speech and clinical subjective ratings of severity of depression" [39]. Correspondingly, the literature proposes a large number of depression markers in speech signals [8, 19, 21, 23, 24, 41, 48, 51, 56, 59]. In many cases, the markers were identified by comparing the values of a feature in depressed and non-depressed speakers or in speakers affected by different levels of depression severity. The seminal experiments in [8] show that there is a statistically significant correlation between the speaking rate (number of words per minute) and the severity of depression as self-rated by 7 patients. The same work suggests that depression is also likely to increase number and length of pauses and variation of the fundamental frequency. Another pioneering study [19] involved 115 speakers and the analysis was performed separately for female and male speakers. In both cases, it was shown that formants and Power Spectral Density are the most discriminative features for distinguishing between depressed, dysthimic or suicidal speakers and control ones. Similar experiments in [41] show that the glottal flow spectrum allows one to discriminate between control, depressed and suicidal speakers (the experiments involved 10 individuals). Finally, the experiments in [56], performed over 9 speakers, show that depressed and non-depressed individuals differ in terms of jitter (variability of the period duration of the speech signal) and shimmer (variations in the amplitude of the vocal cord vibration), two measurements related to voice quality.

More recent work focuses on the analysis of *landmarks* i.e., events that "specify points in *time* for different abrupt articulatory events" [23]. These include, e.g., start and end of sustained vibration of vocal folds periods, start and end of sustained periodicity, etc. Count and duration of the events and of their N-grams (sequences of N consecutive events) discriminate depressed and non-depressed speakers with F1 Scores between 30% and 40% over the DAIC-WOZ Corpus [21, 24]. Other experiments revolve around the *vowel space*, defined as "the frequency range spanned by the first and second formant of the vowels /i/ [...] /a/ [...] and /u/ [...] with respect to the reference sample" [48]. The results of the analysis, performed over 253 speakers, suggest that the vowels uttered by depressed individuals tend to span reduced frequency ranges, thus allowing one to discriminate effectively between depressed and control speakers. In addition, measurements related to the range were shown to be robust with respect to gender, ethnicity and education.

Another approach commonly used to identify depression traces is to test whether enriching an existing feature set with measurements accounting for a particular acoustic phenomenon (e.g., jitter) improves the performance of a depression detector. This is the case in [51], where the experiments involve 110 speakers (including 56 diagnosed with depression by professional psychiatrists) and show that speaking rate and pauses (number and total length) can decrease the error rate of a depression detection approach by roughly 50%. The experiments in [59], performed over the AVEC dataset, use a similar approach and show that the structure of the correlations between values of formant frequencies and delta-mel-cepstra extracted at different points in time can improve the performance of a depression detector.

Overall, the results presented in this section suggest that there are detectable differences between depressed and non-depressed speech. However, to the best of our knowledge, no attempt was made to investigate whether depression can be detected in any point of a speech signal (depression traces are continuous) or only at specific points in time (depression traces are punctual). Furthermore, no attempt was made to investigate such problem with embeddings (see, e.g., [5]), i.e., with recently developed speech measurements that represent speech effectively, but are not interpretable like the other measurements considered in this section. The goal of this work is to bridge, at least to an initial extent, such a gap.

2.2 Depression Traces in Language

Like in the case of speech signals, there are "psychological theories that lend support for the existence of certain linguistic features of depressed language" [39]. Correspondingly, several works investigated the measurable traces that depression leaves in language [7, 15, 31, 44, 47, 54, 55]. One of the first approaches was to identify words that depressed speakers tend to use more frequently. The analysis in [47], based on the Linguistic Inquiry Word Count [53], showed that depression patients tend to use more frequently first person pronouns and negatively valenced words, while using less frequently positively valenced words. Such results, obtained over the essays written by 339 students (253 female), were observed in most other works and are probably among the most commonly observed effects of depression. In a similar vein, the key-goal in [7] is to find groups of words that can account for Behavioral Activation, i.e., the tendency to perform actions that result into reward or gratification. The reason is that depressed individuals are expected to show lower Behavioral Activation and, as a consequence, should use less frequently the words that account for such a construct. The experiments, performed over material posted by 10,718 persons during online chat therapies, confirmed such an expectation. The focus in [44] is on cultural differences between Spanish and English speakers. The analysis is based on the association between LIWC categories and depression. The experiments were performed on 320 texts (160 in English and 160 in Spanish) posted in online depression forums. Overall the results suggest that depression leaves the same traces in both English and Spanish speaking people, in particular both have a tendency to use more frequently first person pronouns.

The goal of the work in [31] is to build a *lexicon*, i.e., a list of words likely to act as depression markers. The experiments were performed over 58625 microblogs posted by depressed individuals (965 in total) and 52787 microblogs posted by non-depressed ones (903 in total). The lexicon was built by identifying a relatively small set of seed words first (less than 100 terms), and then by measuring the similarity of their word2vec embeddings [37] with the embeddings of all other words in the data. The effectiveness of the lexicon was measured by using the presence of the lexicon words as a feature for depression detection. The results show F1 Scores up to 89% with a lexicon of 21321 words (e.g., *collapse*, *stress*, *suicide*, etc.). A similar approach was proposed in a work aimed at crafting "[...] alternative lexicons that contain more informal language and thus are more linguistically appropriate for text message classification" [54]. The 36 lexicons were built with Empath [17] and 14 of them allowed to perform depression detection above chance.

Most works described in this section make use of texts in which people talk about depression and this might make it easier to detect traces of the pathology. For this reason, the experiments in [60] revolve around texts in which depressed people write about different topics. The rationale is to obtain more realistic estimates of text-based depression detection performance. The experiments were performed over Reddit posts written by 4324 depressed users and 7153 non-depressed users (roughly 1.5×10^8 words). The results show that an approach fed with LIWC features and N-grams (frequency of N words sequences) achieves F1 Scores up to 68.1% when testing over data about topics different from depression. The use of classification approaches to investigate whether linguistic features actually convey depression-relevant features was used in [55] too, where a feature set based on LIWC and hand-crafted features accounting for the presence of words associated to depression (e.g., "my depression", "my anxiety", etc.) were shown to achieve F1 Scores up to 73% (in conjunction with word embeddings).

A similar approach was used in [33], where LIWC-based features designed to capture sentiment were shown to predict effectively the presence of depression in text messages. Finally, the experiments in [15] were performed over Facebook posts (683 people) and show that there is a statistically significant association between multiple LIWC categories (e.g., negative emotions, first person pronouns, etc.) and depression. Furthermore, the experiments showed that the observed frequency of 200 topics extracted automatically through Latent Dirichlet Allocation [6] allows one to predict whether someone will be diagnosed with depression with Area Under the Curve up to 72%., especially when using Facebook posts collected in the six months preceding the actual diagnosis.

Overall, this brief state-of-the-art gives two main indications. The first is that LIWC appears to be the tool most commonly used to analyze the traces of depression in language (it is used in virtually all works in the area). The second is that no previous work investigated the problem of whether the traces are continuous or punctual. For this reason, the experiments of this work use LIWC to represent the transcriptions of the speech signals and try to test whether depression traces are continuous or punctual.

3 The Approach

Figure 1 shows the main steps of the approach used in the experiments. The first step is the *segmentation* (see Section 3.1), followed by *feature extraction* (see Section 3.2), *classification* (see Section 3.3 for speech signals and Section 3.4 for language) and, finally, *aggregation* (see Section 3.5). The approach is applied in parallel to speech signals and their transcriptions. The unimodal outcomes are then combined through late fusion (see Section 3.6).

3.1 Segmentation

The *segmentation* step splits the input recordings and their respective transcriptions into non-overlapping segments. It first splits every original recording into two equal parts, then it splits each of these two into two equal parts and so on, until the number of segments is $N_{max} = 64$. The reason for stopping at such value is that further splitting leads to segments that are too short to be processed automatically.

3.2 Feature Extraction

The feature extraction step converts every segment, whether extracted from the speech signals or from their transcriptions, into sequences of feature vectors (see Figure 1). In the case of the speech signals, the process takes place in two different ways. The first is the extraction of embeddings through the application of *wav2vec2.0*[1] [5], one of the most common speech Foundation Models. The embeddings are extracted from analysis windows of length 25 ms starting at regular time steps of 10 ms (both values are standard in the literature). The motivation behind the use of wav2vec2.0 is that its embeddings lead to state-of-the-art performances in a large number of speech-based tasks aimed at the inference of social and psychological phenomena (see, e.g., [32] for personality traits and [10] for emotions). The dimensionality of the embeddings is $D = 1024$.

The second approach for speech signals is the extraction of handcrafted features from 25 ms long analysis windows starting at regular time steps of 10 ms (see above for the choice of the values). The handcrafted features are extracted with *OpenSMILE* [16] and correspond to the *IS09* set originally designed for the *Interspeech 2009 Emotion Challenge* [49]. The set includes Mel Frequency Cepstral Coefficients 1 to 12 (12 features), Root Mean Square Energy (1 feature), Fundamental Frequency (1 feature), Voicing Probability (1 feature) and Zero Crossing Rate (1 feature). The set is enriched with the delta coefficients for each of the features, thus leading to a vector of dimension $D = 32$. The sequence of vectors resulting from the feature extraction process is denoted in both cases with $X = (\vec{x}_1, \vec{x}_2, \dots, \vec{x}_T)$, where T is the total number of vectors per segment (see Figure 1).

For the transcriptions (see lower part of Figure 1), the feature extraction process converts every segment k into a vector $\vec{z}_k$ by applying the *Linguistic Inquiry Word Count* (LIWC)[2]. Every vector has dimension $D = 94$ and every feature corresponds to the

[1]https://huggingface.co/facebook/wav2vec2-large, last accessed on March 7, 2025.
[2]https://www.liwc.app, last accessed on March 20th, 2025.

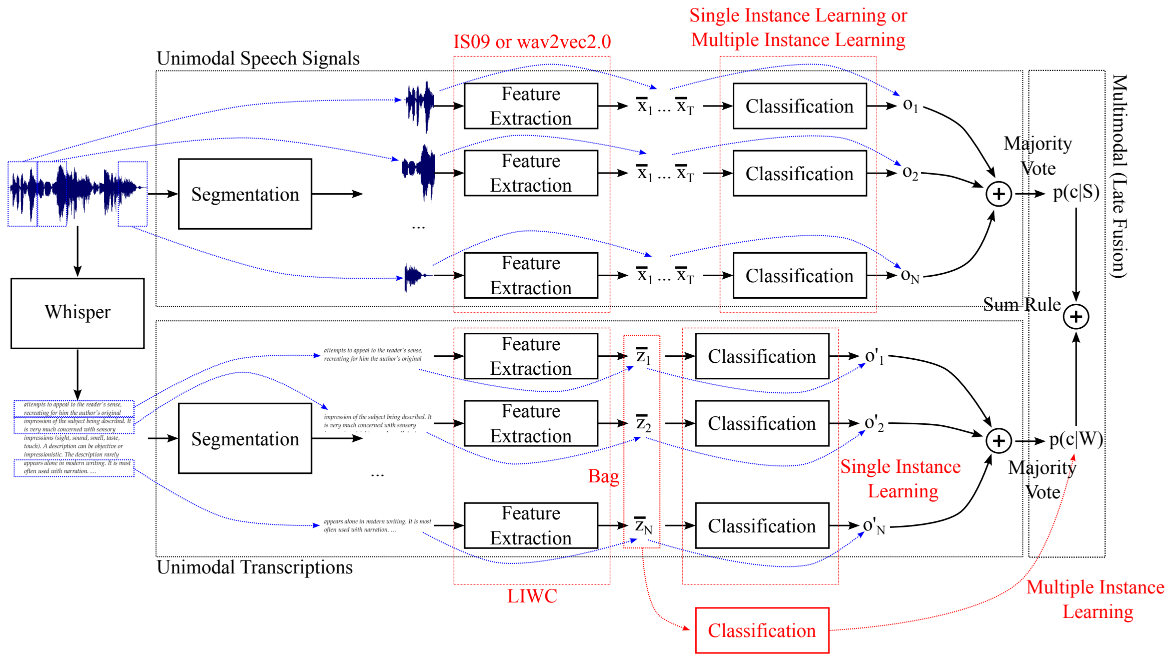

Figure 1: The proposed approach segments the speech data into slices and then classifies individually each of these as belonging to either class Depression or class Control. The individual classification outcomes are then aggregated through a majority vote.

percentage of words that belong to one of 94 manually identified psychologically-relevant categories like, e.g., positive emotions, personal concerns, first person pronouns, etc. (see [53] for the full list of the categories). The main reason behind the choice of the LIWC is that it was shown to effectively represent the social and psychological content of texts in a wide spectrum of contexts (according to the website in footnote 2, the LIWC was used in 22,000 scientific articles). Furthermore, LIWC works even when a text includes only one word. This is useful when the number N of slices increases and, correspondingly, their average length decreases. Finally, Section 2 shows that LIWC is used in virtually every work aimed at identifying the traces of depression in language.

3.3 Speech Signals Classification

The goal of this step is to assign every segment of a speech signal to one of the possible two classes, namely *Depression* (the positive class) and *Control* (the negative class). In the case of speech, the classification is performed in two different ways (see Figure 1). The first, referred to as *Single Instance Learning* (SIL), feeds each vector $\vec{x}_k \in X$ (see previous section) to a Feed Forward Neural Network (FFNN) that gives as output the posterior of class Depression. If such probability is greater than 0.5, the vector is assigned to class Depression, otherwise it is assigned to class Control. The slice is then assigned to class $\hat{c} = \arg\max_{c \in C} n(c)$, where C is the set of the possible classes and $n(c)$ is the number of vectors $\vec{x}_k \in X$ assigned to class c. The assumption underlying the SIL approach is that depression traces are continuous in time and, therefore, the majority of the vectors in a slice should belong to class Depression when a speaker is affected by the pathology.

The second classification approach is referred to as *Multiple Instance Learning* (MIL) and its underlying assumption is that the labeling is *weak*. In other words, even if a segment belongs to class c, that does not necessarily mean that all the vectors extracted from it belong to the same class c. Therefore, the segment should be assigned to class c when a sufficient number of $\vec{x}_k$ vectors are assigned to that class and not when the majority of the vectors are assigned to it (like it happens with the SIL approach). This means that the MIL approach should work better than the SIL one if the depression traces are punctual. When using the MIL, the feature vectors $\vec{x}_k$ extracted from a slice are treated as a *bag of instances*. The bag is input into a Multiple Instance Neural Network (MINN) [58]. Such a model is based on an FFNN with H hidden layers ($H = 3$). For every $\vec{x}_k$, there is an output $\vec{x}_k^\ell$ of the last hidden layer. The set of all $\vec{x}_k^\ell$, where $k = 1, \ldots, T$, is given as input to a MIL pooling layer that provides, for every component j, the following representation:

$$x_j = \max_{k=1,\ldots,T} x_{kj}^\ell, \tag{1}$$

where x_{kj}^ℓ is the j^{th} component of x_k^ℓ and x_j is the j^{th} component of the layer's output. The resulting representation is sent as input to a *softmax* layer that outputs the posterior probability of class Depression.

The training was performed according to a k-fold protocol (k = 5), and the folds were the same as those provided with the Androids Corpus distribution (see Section 4.1). The protocol is person-independent and it ensures that speakers from the training set do not appear in the test set. This approach aims to detect depression rather than simply recognizing individual speakers. The FFNNs used in the experiments are composed of three hidden layers with

32, 64 and 128 neurons, respectively (the activation function is ReLU). Number of layers and number of neurons per layer were selected through cross-validation. The number of training epochs was set to $E = 30$ and the training was performed using the Adam optimizer [29] with binary cross-entropy as a loss function [12]. Given that the networks require a random initialization, the experiments were repeated $R = 5$ times, and the weights were initialized differently at each repetition. For this reason, all performance metrics are reported as averages and standard deviations over the repetitions (see Section 4). All experiments were run using an NVIDIA GeForce GPU and the latest version of the Tensorflow-Keras framework[3].

3.4 Transcriptions Classification

For what concerns the transcriptions (see lower part of Figure 1), the classification is performed in the same way as above. In the SIL case, the vectors $\vec{z}_k$ (one per slice) are fed to an FFNN that gives as output the posterior of class Depression. In the MIL case, the N $\vec{z}_k$ vectors are treated like a bag and a MINN (see above) is used to assign the speaker to one of the two classes. In the latter case, there is no need to perform an aggregation step (see next section).

The training process involved a cross-validation process, using the k-fold ($k = 5$) protocol accompanying the Androids Corpus distribution (see Section 4.1) to ensure consistency with the paralanguage unimodal approach and reproducibility of the results. The cross-validation allowed the selection of two hyperparameters, namely the number of layers and the number of neurons per layer in the FFNNs used for the classification (both SIL and MIL). The networks employed the ReLU activation function and the number of training epochs was set to $E = 30$. The training process used the Adam optimizer [29] and the binary cross-entropy [12] as the loss function. To evaluate the effect of the random network initialization, the experiments were performed $R = 5$ times, each time with different weights as a starting point. Therefore, all performance metrics are presented as averages and standard deviations across these repetitions.

3.5 Aggregation

At the end of the slice classification step, every segment has been classified and the recording can be assigned to the class its segments are most frequently assigned to. In other words, the recording is classified through a majority vote over the individual segments. The only exception is the use of MIL for the slice transcriptions (see previous section). In such a case, the N $\vec{z}_k$ vectors are treated like a bag and the approach classifies this latter as a whole. In any case, the output is still the posterior of class Depression.

3.6 Multimodal Combination

Figure 1 shows that, with the only exception of the MIL approach for the transcriptions, the output of the classification step is a set $O = \{o_1, \ldots, _N\}$ of classification outcomes, where $o_k \in C$ (C is the set of the classes) and N is the number of slices. In the case of speech (see upper part of Figure 1), this allows one to estimate the posterior probability of a class c as $p(c|X) = n(o_k = c)/N$ where $n(o_k = c)$ is the number of times the classification outcome for a slice is c. In the case of the transcriptions (see lower part of Figure 1), the same method can be used when the classification is performed with the SIL approach. Otherwise, the probability $p(c|W)$ is estimated by the MINN that uses the set of the $\vec{z}_k$ as a bag (see red dotted path in Figure 1).

In all cases, the outcome of the classification process is a posterior probability and this makes it possible to apply the sum rule [30]:

$$\hat{c} = \arg\max_{c \in C} [p(c|X) + p(c|W)], \tag{2}$$

where C is the set of the classes.

[3] https://www.tensorflow.org, last accessed on March 20th, 2025.

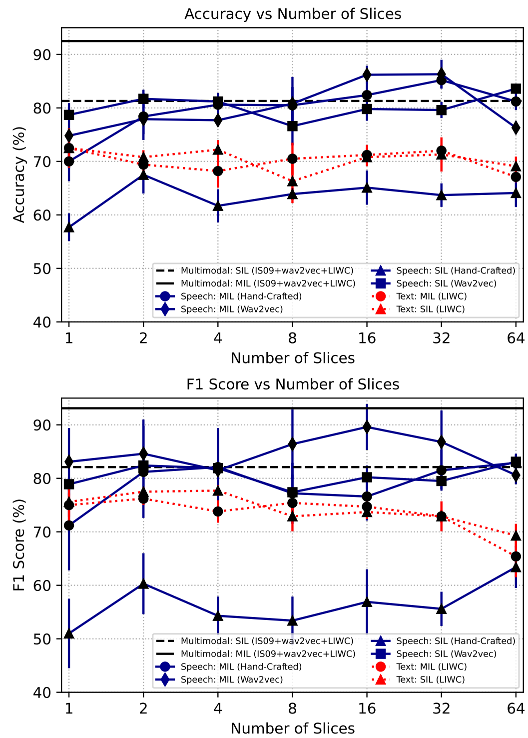

Figure 2: The horizontal axis is in logarithmic scale and shows $\log_2 N$, where N is the number of segments.

Ref.	Accuracy (%)	F1 Score (%)
Baseline [52]*	83.9±1.3	84.7±0.9
This work	92.5±1.1	93.1±1.1
[14]	×	92.0
[26]	95.3±4.2	95.8±3.8
[27]	87.0±7.4	86.7±6.6
[40]	87.0±1.2	88.9±1.7
[42]	87.9	83.3
[43]	79.8±7.4	80.6±7.8
[61]	88.2±8.5	86.3±11.5

Table 1: All results of this table were obtained over the data of the Androids Corpus. Not all works mentions explicitly whether they use the folds provided with data, but the values still provide a good reference for comparison. The symbol "×" means that the metric is not reported in the work.

Table 2: The table provides demographic information about the participants. Acronyms F and M stand for Female and Male, respectively.

Group	F	M	Avg. Age	Age-Range
Depressed	43	21	47.9±12.4	23-71
Control	41	11	48.0±12.4	19-68
Total	84	32	47.9±12.3	19-71

4 Experiments and Results

This section describes the Androids Corpus (see Section 4.1) and then reports on the results obtained during the experiments (see Section 4.2).

4.1 The Data

The experiments of this work were performed over the spontaneous speech samples of the Androids Corpus, a publicly available collection of 116 interviews recorded in different mental health centers in Southern Italy [52][4]. The interviews were recorded in the same rooms where doctors and patients meet for their therapeutic sessions. This ensures that the depression patients are familiar with the setting and, furthermore, it reproduces the normal working conditions of the doctors. This follows the recommendations in [22] that data should be collected in conditions as realistic as possible to assess the potential of depression detection technologies outside the laboratory. The interviews were recorded with the microphone of a laptop placed always in the same position between participants and interviewers. The goals are to limit the costs of the recording approach (no need for expensive microphones or other devices) and to avoid invasive sensors [25] that might be difficult to accept for depression patients. Every interview included the same questions about everyday life (e.g., "What did you do last week end?", "Tell me about a movie you liked", etc.) that were posed always in the same order.

Collecting data during interviews is not the only option in depression detection. Other works make use of read speech [51] or monologues (e.g., people describing pictures without interacting with others) [34]. However, most works using both read and spontaneous speech suggest that this latter type of data, especially when recorded during interviews, tends to lead to better results [2, 35, 38]. For this reason, the experiments of this work were performed over interview recordings.

The 116 participants, one per interview, include 52 persons who never experienced mental health issues (referred to as *control* participants) and 64 individuals that were diagnosed with depression by professional psychiatrists. The diagnosis was performed according to the latest version of the *Diagnostic and Statistical Manual of Mental Disorders* (DSM-V) [57]. Table 2 provides demographic information and shows that gender and age distributions are the same, within statistical fluctuations, for both control and depressed participants. This ensures that any observed difference depends mostly on depression and not on other factors. The number of female depressed participants is 2.9 greater because, according to epidemiological observations, women tend to experience depression roughly two times more frequently than men [46]. Furthermore, the age distribution matches the age-range in which it is most common to develop depression [20]. In this way, the 116 participants are likely to reproduce an average sample of the general population. All participants were involved on a voluntary basis and the data was collected according to the regulations of the country where the data were collected (see [52] for more details).

The average duration of the recordings is 184.9±79.4 seconds. If considering separately depressed and control participants, the averages are 173.4±94.4 and 200.5±49.1 seconds, respectively. Such a difference is not statistically significant according to a two-tailed t-test.

[4]https://github.com/androidscorpus/data, last accessed on March 7, 2025.

4.2 Depression Detection Results

The first step of the experiments is to ensure that the approach used in this work (see Section 3) achieves state-of-the-art performances. For such reason, all experiments presented in this section follow the experimental protocol provided with the Androids Corpus distribution, namely a k-fold cross-validation with $k = 5$ (The Corpus includes the folds so that the protocol can be reproduced exactly) [52]. The protocol is *person-independent*, i.e., a speaker is never represented in both training and test set. In this way, it is possible to ensure that the approach recognizes depression and not the voice of the speakers. Table 1 shows the results obtained over the Androids Corpus in this work and in other works of the literature. Not all works mention explicitly whether they used exactly the same protocol as the one used in this work. Furthermore, not all works mention whether their results were obtained over read or spontaneous speech (only this latter was used in this work). However, the table suggests that the results of this work are in line with the state-of-the-art and improve, to a statistically significant extent, over the baselines provided with the Corpus ($p < 0.01$ according to a two-tailed t-test). This is important because it shows that the approach used in this work is as effective as the others proposed in the literature, at least for what concerns the methodologies applied to the Androids Corpus.

The key-goal of the experiments is to understand whether physical, machine detectable traces of depression are punctual (they appear only at specific points in time) or continuous (they appear at any moment in speech). The problem is addressed by splitting the data into shorter and shorter segments that are classified individually. The aggregation of the segment-level classifications is then performed through a majority vote (see Section 3). The rationale is as follows: if the traces are continuous, the classification performance should not depend on the number of segments because the depression traces should be available and detectable in every segment. Vice versa, if the traces are punctual, the classification performance should decrease when the number of segments increases. The reason is that less and less segments will contain depression traces and, therefore, the majority vote will fail in discriminating between depressed and non-depressed speakers.

The classification is performed with methodologies based either on Single Instance Learning (SIL) or on Multiple Instance Learning (MIL). The comparison between the two can shed further light on the question whether the depression traces are continuous or punctual. In fact, MIL should significantly outperform SIL in case the traces are punctual. The reason is that MIL-based approaches can assign a segment to class Depression even when a few of the vectors extracted from the segment belong to class Depression. Vice versa, SIL-based approaches classify a segment to class Depression only if a majority of the vectors extracted from the segment are assigned to such class. Therefore, SIL-based approaches will be less effective than MIL-based ones if the traces are punctual.

Figure 2 shows the results of the comparison for accuracy (upper plot) and F1-Score (lower plot). All following considerations will make use of the F1-Score because such metric is more suitable for unbalanced data like those of the Androids Corpus. In the case of the speech signals (the blue continuous lines in the figure), the performance of all approaches remains stable when the number N of segments increases. In several cases, the segmentation appears to be even beneficial: the SIL approach based on the IS09 features achieves its best F1 Score when $N = 64$ and the performance is higher, to a statistically significant extent, than when $N = 1$ (F1 Score 51.0±6.5% for $N = 1$ vs F1 Score 63.4±3.9% when $N = 64$, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction). The same applies to the SIL-based approach that uses wav2vec2.0 (F1 Score 78.9±1.2% for $N = 1$ vs F1 Score 83.1±1.2% when $N = 64$, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction). In the case of the MIL-based approaches, the best F1-Score is obtained when $N = 64$ when using the IS09 features (F1 Score 71.2±8.4% for $N = 1$ vs F1 Score 82.9±1.7% when $N = 64$, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction) and when $N = 16$ when using wav2vec2.0 (F1 Score 83.1±6.3% for $N = 1$ vs F1 Score 89.6±4.3% when $N = 64$, $p < 0.05$ according to a two-tailed t-test).

Overall, the observations above suggest that the traces are continuous because the performance does not decrease when N gets greater. However, there is a difference between IS09 features and embeddings. In the first case, MIL-based approaches significantly outperform the SIL-based ones (F1 Score 63.4±3.9% for $N = 64$ for SIL vs F1 Score 82.9±1.7% when $N = 64$ for MIL, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction), while in the second case, there is no statistically significant difference between the two (F1 Score 83.1±1.2% for $N = 64$ for SIL vs F1 Score 89.6±4.3% when $N = 16$ for MIL, $p > 0.05$ according to a two-tailed t-test with Bonferroni Correction). One possible interpretation of these results is that embeddings capture depression traces that are continuous, while IS09 features capture traces that are punctual (hence the better performance of MIL), but still appear frequently enough to make the approach robust to increases in the number of segments.

For what concerns the text-based unimodal approaches, the performance remains roughly the same for both SIL and MIL (no statistically significant differences with respect to $N = 1$) until $N = 32$. It is only for $N = 64$ that there is a statistically significant drop in performance for both SIL (F1 Score 75.6±3.5% for $N = 1$ vs F1 Score 69.3±2.2% when $N = 64$, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction) and MIL (F1 Score 75.0±2.2% for $N = 1$ vs F1 Score 65.4±3.9% when $N = 64$, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction). One possible explanation is that the text segments include less and less words and this damages a representation like LIWC that is based on statistics (every feature is the probability of observing a word belonging to a certain category). In this respect, the results seem to suggest that the depression traces in language are continuous, but they are detectable only as long as the slices are long enough for the LIWC representation. This is further confirmed by the comparison between SIL and MIL. When comparing the best performances obtained with the two approaches, there is no statistically significant difference (F1 Score 77.7±1.0% for $N = 4$ for SIL vs F1 Score 76.2±1.2% when $N = 2$ for MIL, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction).

Besides the unimodal approaches, Figure 2 shows the performances of two multimodal approaches, one based on SIL and the other based on MIL (see dashed and continuous black horizontal lines, respectively). In both cases, the approach was obtained through late fusion (see Figure 1). Furthermore, in both cases the multimodal approach was obtained by combining the unimodal approaches leading to the highest performances with IS09 features, embeddings and LIWC. In the case of SIL, this led to an accuracy of 81.0%±2.1% and an F1 Score of 83.3%±2.1% (see dashed horizontal black lines in Figure 2), while in the case of MIL it led to an accuracy of 92.5%±1.1% and an F1 Score of 93.1%±1.1% (see continuous horizontal black lines in Figure 2). The MIL-based multimodal approach outperforms to a statistically significant extent the best unimodal approach in terms of Accuracy ($p < 10^{-3}$ according to a two-tailed t-test with Bonferroni Correction), but not in terms of F1 Score. One possible explanation is that the text-based approaches tend to achieve lower performances and, therefore, they do not help to improve the multimodal combination.

The MIL-based multimodal approach outperfoms the SIL-based one to a statistically significant extent in both Accuracy and F1 Score ($p < 0.01$ according to a two-tailed t-test in both cases). One possible explanation is that speech signals and their transcriptions jointly convey the presence of depression only punctually. In other words, the two modalities are *diverse* enough to make the multimodal combination effective (meaning, e.g., that they tend to make different mistakes over the same segments [45]). Probably because of this, they agree with each other only punctually. Such an observation explains why the MIL-based approach is more effective than the SIL-based one.

Section 4.1 shows that, for every speaker, the Androids Corpus provides the gender. This makes it possible to test whether the considerations made above apply to both female and male speakers or whether there are differences. Figure 3 shows the variation of the F1 Score with N for female (upper plot) and male (lower plot) speakers separately for both unimodal and multimodal approaches. Overall, the patterns are the same as those observed over the whole Corpus (see Figure 2) and all the considerations made above about depression traces apply in the same way to female and male speakers. However, the best performance achieved over male speakers is statistically significantly higher than the best one observed for female speakers for unimodal approaches based on speech signals (F1 Score 82.9±2.4% for $N = 32$ vs F1 Score 80.5±1.2% when $N = 4$, $p < 0.05$ according to a two-tailed t-test). Vice versa, the performance over male speakers is lower on males for transcriptions (F1 Score 73.6±2.0% for $N = 4$ vs F1 Score 77.4±1.4% when $N = 1$, $p < 0.01$ according to a two-tailed t-test with Bonferroni Correction). In the multimodal approach, the best performance for female speakers is higher than for male ones (F1 Score 93.3±0.9% for $N = 1$ vs F1 Score 90.1±0.1% when $N = 1$, $p < 0.01$ according

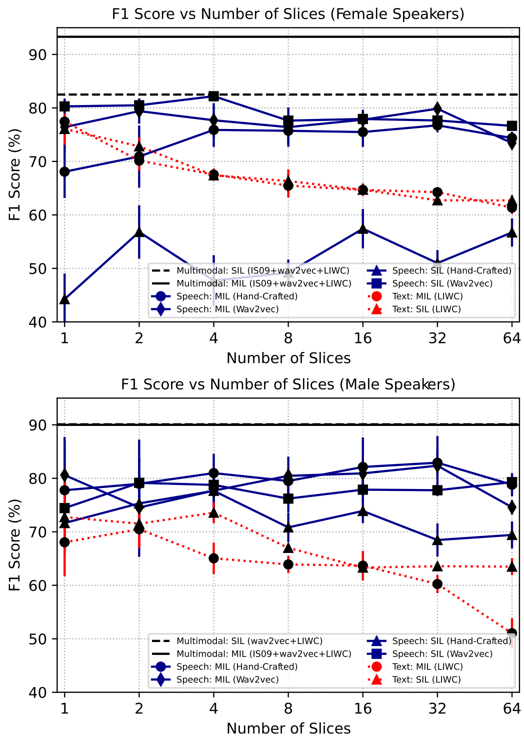

Figure 3: The horizontal axis is in logarithmic scale and shows $\log_2 N$, where N is the number of segments.

to a two-tailed t-test with Bonferroni Correction). This is in line with previous results in the literature [50] and one possible explanation is that female speakers are significantly more than the male ones, mainly due to epidemiological factors (women are more likely than men to suffer from depression [46]). One important observation is that handcrafted features seem to be significantly less effective for female speakers (see the lowest curve in the upper plot of Figure 3). Furthermore, the best result for speech signals in females corresponds to a SIL approach. This seems to suggest that, at least for such channel, women might show depression traces more continuously than men.

5 Conclusions

The main goal of this article is to test whether depression traces are punctual or continuous in speech signals and their transcriptions. The results suggest that, for what concerns speech signals, the answer depends on the representation. When using the IS09 handcrafted features, the traces appear to be punctual because MIL-based approaches significantly outperform SIL-ones. In contrast, when using embeddings extracted with wav2vec2.0, the traces appear to be continuous because MIL and SIL approaches do not differ to a statistically significant extent. However, both representations appear to be robust to the segmentation into progressively shorter slices of the data, thus suggesting that, while being punctual, the traces captured by handcrafted features are still frequent enough to be available in most segments. For what concerns the transcriptions, the traces appear to be continuous because there are no statistically significant differences between SIL and MIL approaches. However, drops in performance are observed when increasing the number N of segments beyond 32. One possible reason is that the number of words per segment keeps decreasing and this damages a representation like LIWC that is based on statistics.

In the case of the multimodal approaches, the results suggest that the traces are punctual, meaning that speech signals and transcriptions jointly convey depression-relevant information only at specific points in time. One possible reason is that the two modalities are diverse and, therefore, they tend to disagree when used to recognize the same data. It is probably thanks to such a characteristic that the multimodal approach outperform the unimodal ones to a statistically significant extent.

One of the main observations of the experiments is that splitting the data into shorter and shorter segments leads to an improvement of the performances for all unimodal approaches. One possible explanation is that the segmentation is an effective form of data augmentation and the most important implication of the result is that depression can be detected effectively even when using short segments, especially with embeddings and MIL methodologies. Such result confirms the Thin Slices Theory [3, 4], a psychological theory suggesting that people convey their inner states through short behavioral displays. Furthermore, the result suggests that a person does not need to speak for long time before depression becomes detectable and this is important in view of clinical applications because depression patients find it challenging to engage in social interactions. The same observation can be made in the case of the multimodal approach, but only when using the MIL approaches. The reason is that the diversity of speech signals and transcriptions tends to make the depression traces more punctual (see above).

To the best of our knowledge, this is the first work that tried to show whether depression traces are punctual or continuous. The main limitation of the work is the use of a limited set of representations. However, the feature sets used for both speech signals and transcriptions are among the most commonly applied. Similarly, the results were obtained using a specific approach and the implications might change when considering other approaches. However, the system used in this work achieves state-of-the-art results over a publicly available benchmark (Table 2 shows that the results are better than most other works using the Androids Corpus). Therefore, approaches like those used in this work are worth further application and investigation. Based on the previous considerations, future work will investigate different data representations and different methodologies for depression detection.

Safe and Responsible Innovation Statement. Depression is a major societal issue and this work aims at its its detection. Furthermore, one of the main implications of this work is that depression can be detected with a limited amount of data (less than 10 seconds), an advantage for depressed people who cannot engage long time in interactions. Finally, the experiments analyze possible gender biases.

Acknowledgments

The work was supported by the UKRI Center for Doctoral Training in Socially Intelligent Artificial Agents (EP/S02266X/1).

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